

Meng Zhang

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EXPERIENCE

[Shopee\(E-Commerce\)](#) Singapore Expert Engineer Jun 2022 ~ Present

- Cold startup time optimization:
 - Reduced p80 cold startup time for Shopee (Southeast Asia's largest e-commerce platform with over 20 million DAUs) from **7.6s** to **4.5s** by implementing parallel execution of startup tasks, multi-level image caching, and proactive image preloading, integrating Baseline Profiles, requiring extensive upgrades to multiple Gradle plugins, and automating Baseline Profile generation across different A/B test configurations. Grayscaled by regions.
 - Additional optimizations included raising Main and Render Thread priority, Dex2Oat enhancements, and leveraging prefetching and preloading techniques.
- App fluency optimization:
 - Improved block rate by **14%** through asynchronous image loading using Glide, optimizing ImageView placeholders for faster rendering.
- Chat optimization:
 - Improved message delivery success rate (P80) from **94%** to **99%** by implementing network optimization, user token expiration validation, and migrating from TCP to HTTP requests.
 - Reduced chat conversation list P80 load time from **1.35s** to **800ms**, and decreased chat room P80 load time from **926ms** to **600ms**.
- Product and Communication Leadership:
 - Led the dynamic configuration development of the bottom navigation bar for Shopee's homepage, enhancing flexibility and user experience.
 - Managed overall communication within the Chat team, overseeing the development and iteration of new message types, ensuring seamless feature rollouts and team coordination.

[KuaiShou\(Short Video\)](#) Beijing Expert Engineer Dec 2018 ~ Jun 2022

- ANR optimization:
 - Reduced ANR rate for Kwai (Brazil's largest TikTok competitor with over 40 million DAUs) from **1.2%** to **0.7%** by resolving Broadcast, Service, and Input timeout issues.

- Cold startup time optimization:
 - Improved p50 cold startup time for KuaiShou (China's second-largest short video platform with over 300 million DAUs) by **28%**, reducing it from **5.6s** to **4.0s** through time-cost optimizations and continuous lab monitoring.
- Others:
 - Customized Flutter engine builds, reducing the Android engine size by **85.7%** (from 3.5M to 0.5M) via dynamic loading of libapp.so; a related [PR](#) was accepted by the Flutter team.
 - Improved [Tinker](#) success rate from **80% to 94%**, enabling android developers to fix online bugs.
 - Modularized KuaiShou's app code, significantly accelerating packaging speed and enhancing development efficiency for over 200 Android developers.

[Chain GrowUp\(Startup\)](#) Beijing Android Team Leader Mar 2018 ~ Nov 2018

- Full-Cycle App Development:
 - Led a team of 2 through continuous iteration, handling development, testing, and successful launch of two apps from scratch.
 - Developed [FbeePay](#), an Ethereum wallet with support for batch transactions and transaction cancellations, built entirely from the ground up
 - Released [Fbee](#), a blockchain community app where users can create communities, engage with fans, and conduct token airdrops.

[Meituan\(O2O\)](#) Beijing Senior Android Engineer Jul 2015 ~ Mar 2018

- Hotfix SDK Development ([Robust](#)):
 - Led the development of the [Robust SDK](#), achieving a **99.9%** success rate for hotfixes, enabling developers to fix online bugs without requiring a new app release.
 - Designed Robust to insert code into nearly every method and generate online-runnable binary code automatically while adhering to ProGuard rules.
 - The project was open-sourced on GitHub, accumulating over **4k** stars, and accompanied by two articles published on the Meituan Technology Blog: [Android hot patch](#) and [Android hot patch solution Robust open source,add automated patch tools](#), discussing automated patching tools.
- SDK Development for Meituan:
 - Contributed to the development of company-wide SDKs including UUID (Uniform Device Identifier), a server-generated identifier shared across all Meituan apps.
 - Developed the Passport SDK, providing unified functionality for user sign-up, login, and logout across all Meituan applications.

EDUCATION

2011.09~2015.06 Xi'an Jiaotong University Software Engineering Bachelor